Question	Answer	Marks
6(a)	joule / coulomb	**B1**
6(b)(i)	lamps have same p.d./lamp<u>s</u> have p.d. of 2.7 V	**B1**
	current = 0.15 + 0.090 = 0.24 A	**A1**
6(b)(ii)	$R = (4.5 - 2.7) / 0.24$ **or** $R_P = 18\,(\Omega)$ and $R_Q = 30\,(\Omega)$ $I / R_T = 1/18 + 1/30$ and so $R_T = 11.25$ $4.5 = 0.24 \times (R + 11.25)$	**C1**
	$R = 7.5\,\Omega$	**A1**

Question	Answer	Marks
6(b)(iii)	$R = \rho l / A$	C1
	$R_P / R_Q = [(2.7/0.15)/(2.7/0.09)]$ (= 0.60)	C1
	ratio $= 0.60 \times 2^2$ $= 2.4$	A1
6(b)(iv)	less p.d. across resistor/greater p.d. across <u>P</u>	B1
	greater current through <u>P</u> and so resistance (of P) increases	B1

Question	Answer	Marks
7(a)	arrow pointing vertically down the page	B1
7(b)	$E = ½mv^2$	C1